Chapter 5 full solutions

DM05-E01 • One formula with different types

Equality $3a - 7 = 3b - 7$ gives $3(a - b) = 0$, hence $a = b$, proving injectivity. An output y can only have preimage $(y + 7)/3$, an integer iff $y \equiv 2 \pmod 3$. Zero is missed, so the map is not onto integers and has no two-sided inverse of this type.

Changing the codomain to $2 + 3\mathbb{Z}$ makes each output have that unique integer preimage, giving a bijection. Its inverse is $y \mapsto (y + 7)/3$. Direct substitutions give $((3x - 7) + 7)/3 = x$ and $3(y + 7)/3 - 7 = y$. On rationals every rational output also has a rational preimage by the same formula, so $\mathbb{Q} \to \mathbb{Q}$ is bijective.

DM05-E02 • Nonempty fibers and named servers

Start with $S(3, 1) = 1, S(3, 2) = 3, S(3, 3) = 1$. Then

$$S(4, 2) = 1 + 2 \cdot 3 = 7, \quad S(4, 3) = 3 + 3 \cdot 1 = 6, \quad S(5, 3) = 7 + 3 \cdot 6 = 25.$$

These are 25 unlabeled nonempty partitions. Label their blocks by the three servers in $3! = 6$ ways, giving 150 assignments.

Alternatively $3^5 = 243$ total functions, minus three missed-server classes each of size $2^5 = 32$, plus three constant functions missing two servers, gives $243 - 96 + 3 = 150$. A function missing t outputs receives net weight $(1 - 1)^t$, correctly handling overlaps.

Unnamed servers retain only fibers, giving 25. Division by six is valid because each unlabeled partition has exactly six distinct labelings, not because factorial division is universally available.

DM05-E03 • A finite-set equivalence

For $|A| = n$, an injection has n different outputs inside a set of size n, hence fills it. Conversely, an onto map has n nonempty fibers totaling n elements, so each fiber is a singleton. For zero size, the one empty function satisfies both quantified definitions vacuously.

On $\mathbb{N}_0$, $n \mapsto n + 1$ is injective but misses zero. The map $n \mapsto \lfloor n/2 \rfloor$ hits each k at $2k$, but collides at zero and one. Equal cardinality of sets does not make a specified function bijective.

DM05-E04 • Can composition hide collisions?

If the composite is injective, equal f-values give equal inputs. For $y, y' \in f(A)$ with $g(y) = g(y')$, choose their input preimages. Composite injectivity makes the inputs, hence y, y', agree. Conversely cancel restricted g, then f, in an equality of composite values.

Take $A = \{1, 2\}, B = \{a, b, c\}, C = \{0, 1\}$, with $f(1) = a, f(2) = b, g(a) = 0, g(b) = g(c) = 1$. The composite sends one and two to different values. The collision of g involves c, outside the image.

For onto composition with non-onto f, take $A = \{1\}, B = \{a, b\}, C = \{0\}$, $f(1) = a, g(a) = g(b) = 0$. Adding injectivity of g restores the inference: for every b, composite surjectivity gives $g(f(a)) = g(b)$, and cancellation gives $f(a) = b$.

DM05-E05 • What pigeonholes guarantee

There are 22 residue boxes, giving $\lceil 23/22 \rceil = 2$ congruent values. The integers $0, 1, \ldots, 22$ repeat only residue zero, twice, so three are not guaranteed.

If every box has at most $t - 1$ objects, the total is at most $k(t - 1)$. One more forces a box to reach t. Conversely, putting exactly $t - 1$ in each box supplies a counterexample at that lower total. Thus the minimum is $k(t - 1) + 1$, also valid at $t = 1$, where it equals one.

DM05-E06 • Numeric value and bit length

After t updates, the value is $\lfloor N/2^t \rfloor$, by induction using uniqueness of integer quotients. Since $2^{b-1} \leq N < 2^b$, update $b - 1$ leaves at least one and update b gives zero. There are exactly b updates and $b + 1$ loop-condition tests.

With fixed word width, constant-time shifts and zero tests give exactly b updates, but b is bounded by that width, so this is not an unbounded asymptotic family. In a parameterized variable-length model, copying the updated representation with zero as an empty string copies $b - 1, b - 2, \ldots, 1, 0$ bits, totaling $b(b - 1)/2 = \Theta(b^2)$; representing zero by one character adds 1. Retaining the original string and updating a read boundary gives $\Theta(b)$ loop control when boundary updates and emptiness tests are constant time.

These contracts do not automatically include input reading, all allocator costs, or arbitrary-precision arithmetic on enormous indices. The stated representation assumptions matter.

DM05-E07 • The full criterion for a quotient formula

If $n \mid m$, then $a - a' = tm$ is divisible by n, proving $[a]_n = [a']_n$ for equal input classes. Conversely, equality $[0]_m = [m]_m$ requires equal outputs, so $[0]_n = [m]_n$, forcing $n \mid m$.

Every valid map is onto because $[b]_m$ hits $[b]_n$. At $m = n$, it is the identity and injective. For $m > n$ with $n \mid m$, distinct $[0]_m, [n]_m$ both map to zero, so injection fails. Thus injection is equivalent to $m = n$.

For six to three, fibers are $\{[0]_6, [3]_6\}$, $\{[1]_6, [4]_6\}$, and $\{[2]_6, [5]_6\}$.

DM05-E08 • Structural counting with partitions

Given partition $\mathcal{P}$, define $\pi : A \to \mathcal{P}$ by the unique block containing each input. Coverage gives existence and disjointness uniqueness. Nonempty blocks give surjectivity, with each block exactly a fiber. Equal π-values mean membership in the same block.

Choose a k-point image in $\binom{n}{k}$ ways, partition the domain into k nonempty fibers, then label them. This gives $\binom{n}{k} k! S(m, k)$. Every function has unique image and fibers, so there is no duplication. Different sizes form disjoint classes whose sum is n^m.

At $m = 4, n = 3$, the contributions are 3,42,36, totaling 81. At $m = 0$, the one empty function has image size zero and uses $S(0, 0) = 1$. In particular, $m = n = 0$ uses the counting convention $0^0 = 1$. If $n = 0, m > 0$, no total function exists; the sum over $0 \le k \le \min(m, n)$ is 0, matching $0^m = 0$.

DM05-E09 • Identities and operation tables

A specified identity fixes its row and column, counting their intersection once, so $2n - 1$ cells are fixed. The remaining $(n - 1)^2$ cells each have n choices, giving $n^{(n-1)^2}$. Requiring commutativity leaves only the diagonal and upper triangle on $A \setminus \{e\}$, totaling $n(n - 1)/2$ free cells and $n^{n(n-1)/2}$ operations.

Two identities obey $e = e * e' = e'$, so identity choices give disjoint classes. Requiring some identity therefore gives $n \cdot n^{(n-1)^2}$. At three, the three requested answers are 81,27,243. At one they all equal one.

On a set with more than one element, define $a * b = b$. Every e satisfies $e * a = a$, so is a left identity. A right identity would require $a * e = e = a$ for every a, impossible. This does not contradict uniqueness of a two-sided identity. On a singleton $\{e\}$, closure forces $e * e = e$, so e is a two-sided identity and the requested counterexample is impossible.

DM05-E10 • Keep the correct objects in existence proofs

Use eight prefixes $s_0 = 0, s_j = \sum_{i=1}^{j} a_i$ and seven residue boxes. Equal residues at $i < j$ make $s_j - s_i$ a nonempty contiguous sum divisible by seven. The seven original values might fill different boxes; even a collision would concern their difference, not generally a contiguous sum.

For 17 distinct reals, if neither monotone length reached five, ending-length labels (I_j, D_j) would occupy only $\{1, 2, 3, 4\}^2$, sixteen possibilities. Any earlier/later value comparison strictly increases at least one relevant label, so all labels differ, contradiction.

The sixteen-term example is $4, 3, 2, 1, 8, 7, 6, 5, 12, 11, 10, 9, 16, 15, 14, 13$. An increasing subsequence takes at most one from each of four blocks. A decreasing subsequence stays in one block. Both lengths are at most four.

DM05-E11 • Proving binary exponentiation

Initially $rb^e = 1 \cdot a^N = a^N$. At $e = 2t$, updates preserve $r(b^2)^t = rb^{2t}$. At $e = 2t + 1$, they preserve $(rb)(b^2)^t = rb^{2t+1}$. Combining powers uses associativity, not commutativity of all elements. The exponent strictly descends to zero, where the invariant yields $r = a^N$. For $N = 0$, return one directly.

For 2^{11}, start-state triples and updated results are:

e	b	Initial r	Updated r
11	2	1	2
5	4	2	8
2	16	8	8
1	256	8	2048

Every round squares once and each one-bit multiplies into r, so for $N \geq 1$ the exact count is $\lfloor \log_2 N \rfloor + 1 + \text{popcount}(N)$. Here it is $4 + 3 = 7$. Omitting the final unused square reduces it by one.

Operand lengths grow. The result 2^N alone needs $N + 1$ bits, excluding logarithmic bit time even to output it. The algorithm bounds multiplication count; bit cost must sum the costs at their changing operand lengths.

DM05-E12 • Witnesses for growth and averages

For $n \geq 1$, $7n^2 \leq 7n^2 + 2n + 1 \leq 10n^2$, giving witnesses $c_1 = 7, c_2 = 10, N = 1$. If a linear upper bound existed, fixed c, N would give $7n^2 \leq cn$ eventually. Integer $n > \max(N, c/7)$ contradicts that bound, proving failure of Big-O.

For uniform successful positions, expected comparisons are

$$\frac{3}{4}\frac{5+1}{2} + \frac{1}{4} \cdot 5 = \frac{9}{4} + \frac{5}{4} = \frac{7}{2}.$$

Concentrating all successes at the first cell instead gives $(3/4)1 + (1/4)5 = 2$. Equal success probability does not determine mean cost; the conditional position distribution also matters.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 5.

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https://github.com/xinhi6716/computer-science-self-study

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